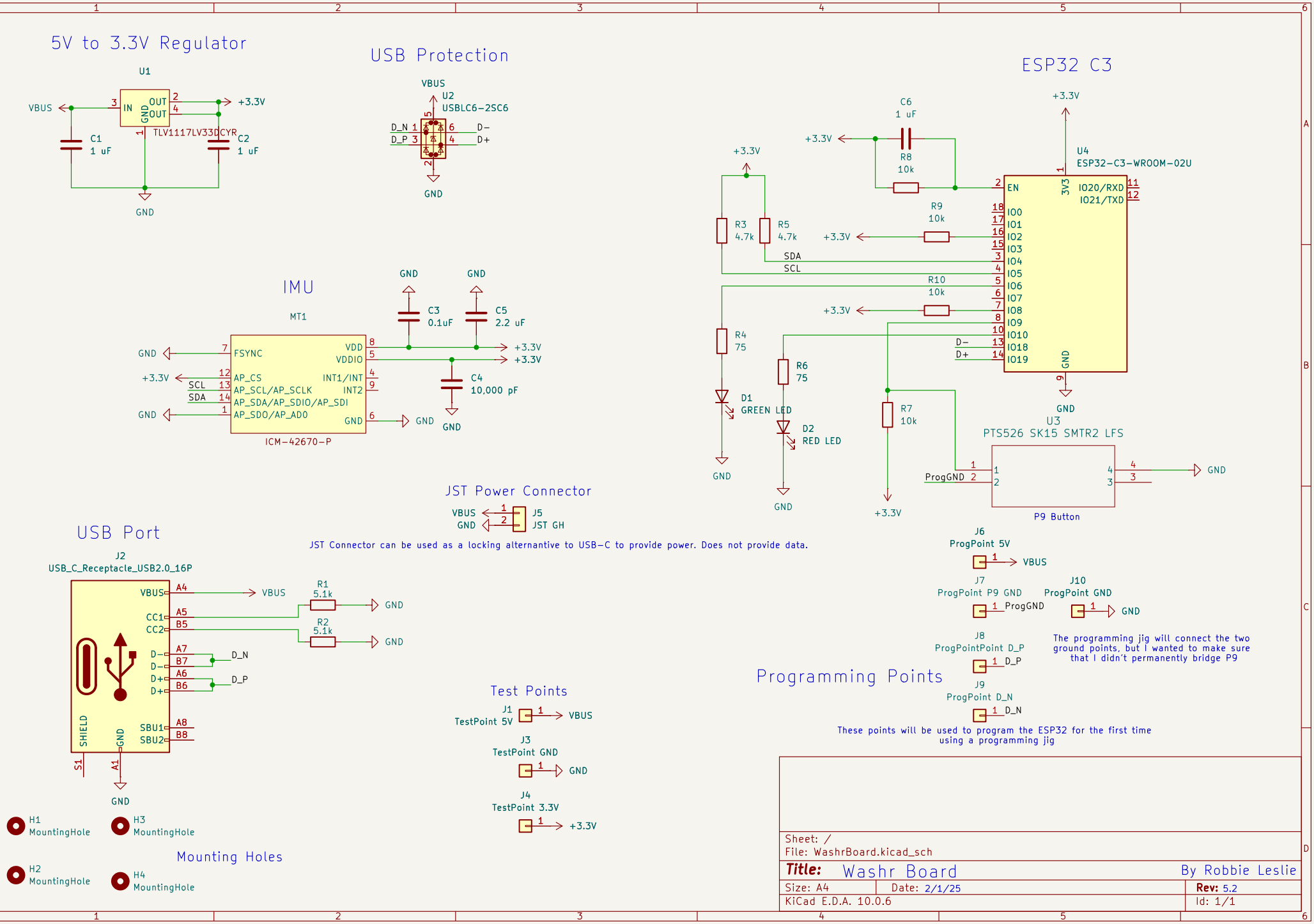

5V to 3.3V Regulator
U1
VBUS
IN
GND
OUT
OUT
+3.3V
TLV1117LV33DCYR
C1
1 uF
C2
1 uF
GND
USB Protection
VBUS
U2
USBLC6-2SC6
D_N
D_P
D-
D+
GND
ESP32 C3
+3.3V
C6
1 uF
R8
10k
U4
ESP32-C3-WROOM-02U
EN
3V3
IO20/RXD
IO21/TXD
R3
4.7k
R5
4.7k
R9
10k
SDA
SCL
IO0
IO1
IO2
IO3
IO4
IO5
IO6
IO7
IO8
IO9
IO10
IO18
IO19
R10
10k
R4
75
R6
75
D1
GREEN LED
D2
RED LED
R7
10k
D-
D+
GND
U3
PTS526 SK15 SMTR2 LFS
ProgGND
P9 Button
IMU
MT1
GND
C3
0.1uF
C5
2.2 uF
C4
10,000 pF
FSYNC
VDD
VDDIO
AP_CS
AP_SCL/AP_SCLK
AP_SDA/AP_SDIO/AP_SDI
AP_SDO/AP_AD0
INT1/INT
INT2
ICM-42670-P
JST Power Connector
VBUS
GND
J5
JST GH
JST Connector can be used as a locking alternantive to USB-C to provide power. Does not provide data.
USB Port
J2
USB_C_Receptacle_USB2.0_16P
VBUS
CC1
CC2
D-
D+
SHIELD
SBU1
SBU2
R1
5.1k
R2
5.1k
D_N
D_P
S1
A1
Test Points
J1
TestPoint 5V
J3
TestPoint GND
J4
TestPoint 3.3V
Programming Points
J6
ProgPoint 5V
J7
ProgPoint P9 GND
J10
ProgPoint GND
J8
ProgPointPoint D_P
J9
ProgPoint D_N
The programming jig will connect the two ground points, but I wanted to make sure that I didn't permanently bridge P9
These points will be used to program the ESP32 for the first time using a programming jig
Mounting Holes
H1
MountingHole
H2
MountingHole
H3
MountingHole
H4
MountingHole
Sheet: /
File: WashrBoard.kicad_sch
Title: Washr Board
By Robbie Leslie
Size: A4
Date: 2/1/25
Rev: 5.2
KiCad E.D.A. 10.0.6
Id: 1/1